

methodological hierarchy for degenerate quantum eigenspaces

	residual PINN	orthogonality penalty	orthogonal projection	subspace Ritz	symmetry sectors
1D excited states	$\pm$	$\pm$		—	—
square degeneracy	×	$\pm$	$\pm$		—
rectangle splitting	?	?	$\pm$		—
disk $m=1$ doublet	?	$\pm$	$\pm$		—
ellipse splitting	?	×	×	$\pm$	
success partial/unstable failure not run/missing not applicable					